Aare Renu Shri

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.30
Sri Chaitanya Junior Girls College, Madhapur, Hyderabad	Class XII (Telangana Board Of Intermediate Education, Hyderabad)	2024	98.40%
Sri Chaitanya Tech HS Sai Mourya Towers, Kodad	Class X (Board of Secondary Education, Telangana)	2022	10.00%

PROJECTS

xv6 OS Kernel Extensions (OS II) | C | Course Project

- Added new system calls in **xv6** for processes to query their own info, backed by kernel-side syscall-count tracking.
- A scheduler with **4** priority levels (**4-level-SC-MLFQ**) is built by using the syscall count, which analyses which processes are interactive, **CPU-intensive** processes, and also makes sure that low priority processes don't get stuck waiting forever.
- Implemented virtual memory using a clock page replacement algorithm, allowing **xv6** to swap pages to disk instead of terminating processes when physical memory is full.
- Extended virtual memory with disk-backed swapping, added **FCFS** and **SSTF** disk scheduling algorithms, and implemented **RAID 0/1/5** to improve storage performance, reliability, and fault tolerance across simulated disks.

Query Optimization & Indexing Analysis | PostgreSQL, Python

- Benchmarked indexing strategies on a **6M-row** TPC-H dataset, measuring **~18% faster** load times when creating an index after bulk data insertion versus before, using isolated table copies and controlled timing experiments.
- Built a Python benchmarking tool measuring query performance across **5 indexing strategies** (no index, index, forced index, clustered index, index-only scan) over **50 selectivity levels**, using **warm-cache** methodology (repeated executions with cache primed) to eliminate disk I/O noise and ensure consistent and fair comparisions.
- Implemented a **recursive query-plan comparator** in Python that parses PostgreSQL's **EXPLAIN (FORMAT JSON)** output and determines structural equivalence between plans, ignoring cost and cardinality noise.
- Designed a **divide-and-conquer algorithm** to detect optimizer plan switches across a parameterized date range, reducing required queries by **~95%** (16 vs. ~365) versus brute-force search, with results independently verified.

Log Classifier API | FastAPI, Docker-ready, Render | Personal Project

- Wrapped an existing trained log classification model (Logistic Regression + TF-IDF, **4 categories**) in a **FastAPI** service with a */classify* endpoint, turning a script-only ML model into something callable over HTTP by any client.
- Identified and fixed a train/inference mismatch by adding the **regex-based log normalization** used during training to the API's prediction path, keeping results consistent with how the model was actually trained.
- Used **Pydantic** to validate incoming request data before it reaches the model, and built interactive API documentation via **Swagger UI**, allowing endpoints to be tested and demoed without writing separate client code.
- Configured a **deployment pipeline** on **Render** with custom build/start commands, **dynamic port** binding via environment variables, and host configuration for external traffic. Set up reproducible deployment using a **virtual environment**, *.gitignore,* and pinned dependencies in *requirements.txt*, resulting in a **live**, publicly accessible endpoint.

SKILLS

Languages: C | C++ | Python (Working Knowledge) | SQL | Flex | RISC-V Assembly

Systems: xv6/UNIX-like OS internals | Basic Linux/Unix(Terminal)

Tools: Git & GitHub | QEMU | gem5 Simulator | Overleaf(Latex)

Other: Digital Fabrication Tools

RELEVANT COURSES

Software Core: Software Development Fundamentals, Data Structures, Algorithms, Database Management Systems.

Systems Core: Operating Systems, Computer Architecture, Compilers.

Mathematics Core: Discrete Mathematics for Computer Science, Probability and Random Variables, Linear Algebra.

POSITIONS OF RESPONSIBILITY

Event Coordinator | DIESTA | IIT Hyderabad

- Monitored squash matches in real time and accurately recorded results across multiple rounds, ensuring all participating teams checked in and started on schedule to keep the overall event running smoothly without delays.
- Coordinated closely with department representatives throughout the tournament to promptly resolve scheduling conflicts and logistical issues, ensuring smooth execution and a positive experience for all participants.

Coordinator, Cultural & Design | CSE Departmental Day | IIT Hyderabad

- Reached out to department alumni across different graduating batches to coordinate a friendly interaction session, gathering their responses and experiences to share with current students during the event.
- Contributed creative ideas for alumni interaction questions and helping brainstorm engaging discussion topics and icebreakers that made the session more interactive, memorable, and enjoyable for both alumni and current students.

Deputy Coordinator | Elan & nVision | IIT Hyderabad

- **Publicity:** Reached out to colleges outside IIT Hyderabad to promote the fest, share event details, and encourage student participation, helping expand outreach and boost overall footfall at the event.
- **Informals:** Helped construct components of a large-scale cardboard showcase installation for the event.

EXTRACURRICULAR

NSS (National Service Scheme), IIT Hyderabad

- Contributed to campus cleanliness drives and participated in walkathons promoting environmental and social awareness.
- Created posters and essays on social causes, and attended awareness talks and events organized by NSS.

Hobbies: Dance, Sketching and Arts